

Incorporation of EPA and DHA into plasma phospholipids in response to different omega-3 fatty acid formulations - a comparative bioavailability study of fish oil vs. krill oil



Background Bioavailability of omega-3 fatty acids (FA) depends on their chemical form. Superior bioavailability has been suggested for phospholipid (PL) bound omega-3 FA in krill oil, but identical doses of different chemical forms have not been compared. Methods In a double-blinded crossover trial, we compared the uptake of three EPA+DHA formulations derived from fish oil (re-esterified triacylglycerides [rTAG], ethyl-esters [EE]) and krill oil (mainly PL). Changes of the FA compositions in plasma PL were used as a proxy for bioavailability. Twelve healthy young men (mean age 31 y) were randomized to 1680 mg EPA+DHA given either as rTAG, EE or krill oil. FA levels in plasma PL were analyzed pre-dose and 2, 4, 6, 8, 24, 48, and 72 h after capsule ingestion. Additionally, the proportion of free EPA and DHA in the applied supplements was analyzed. Results The highest incorporation of EPA+DHA into plasma PL was provoked by krill oil (mean AUC_{0-72 h}: $80.03 \pm 34.71\% \cdot h$), followed by fish oil rTAG (mean AUC_{0-72 h}: $59.78 \pm 36.75\% \cdot h$) and EE (mean AUC_{0-72 h}: $47.53 \pm 38.42\% \cdot h$). Due to high standard deviation values, there were no significant differences for DHA and the sum of EPA+DHA levels between the three treatments. However, a trend ($p = 0.057$) was observed for the differences in EPA bioavailability. Statistical pair-wise group comparison's revealed a trend ($p = 0.086$) between rTAG and krill oil. FA analysis of the supplements showed that the krill oil sample contained 22% of the total EPA amount as free EPA and 21% of the total DHA amount as free DHA, while the two fish oil samples did not contain any free FA. Conclusion Further studies with a larger sample size carried out over a longer period are needed to substantiate our findings and to determine differences in EPA+DHA bioavailability between three common chemical forms of LC n-3 FA (rTAG, EE and krill oil). The unexpected high content of free EPA and DHA in krill oil, which might have a significant influence on the availability of EPA+DHA from krill oil, should be investigated in more depth and taken into consideration in future trials.